It is then obvious to want to be able to work around the model and have fine levels of control over these aspects. TensorFlow's core API, which is the lowest-level, helps you achieve this fine control. Other higher-level APIs are built on top of this very core. The higher the level of the API, the easier it is to perform repetitive tasks and to keep the flow consistent between multiple users.

MNIST is 'Hello World' to machine learning

The Mixed National Institute of Standards and Technology (MNIST) database is the computer vision data set that is used to train the machine learning system. It is basically a set of handwritten digits that the system has to learn and identify by the corresponding label. The accuracy of your model will depend on the intensity of your training. The broader the training data set, the better the accuracy of your model.

One example is the Softmax Regression model, which exploits the concept of probability to decipher a given image. As every image in MNIST is a handwritten digit between zero and nine, the image you are analysing can be only one of the ten digits. Based on this understanding, the principle of Softmax Regression allots a certain probability of being a particular number, to every image under test.

Smart handling of resources

As this process might involve a good bit of heavy lifting, just like other compute-heavy operations, TensorFlow offloads the heavy lifting outside the Python environment. As the developers describe it, instead of running a single expensive operation independently from Python, TensorFlow lets you describe a graph of interacting operations that run entirely outside Python.

About TensorFlow
Developed by: Google Brain Team
Last stable release: 1.0
Repository: GitHub
Written in: Python and C++
Supports: Linux, Mac and Windows
Licence: Apache 2.0
Website: wwwtensorflow.org

A few noteworthy features

Using TensorFlow to train your system comes with a few added benefits.

Visualising learning: No matter what you hear or read, it is only when you visually see something that the concept stays in your mind. The easiest way to understand the computational graph is, of course, to understand it pictorially. A utility called TensorBoard can display this very picture. The representation is very similar to a flow or a block diagram.

Graph visualisation: Computational graphs are complicated and not easy to view or comprehend. The graph visualisation feature of TensorBoard helps you understand and debug the graphs easily. You can zoom in or out, click on blocks to check their internals, check how data is flowing from one block to another, and so on. Name your scopes as clearly as possible in order to visualise better. END

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